MTH212M Quiz Part A (February 6, 2026)

Name:

Roll No.:

To be submitted: 7:15 pm (+15 minutes for DAP students)

Question 1. ($2+2+2+2$ marks) Prove or disprove the following statements. Disproving may be done via counter-examples.

(a) If two states in a Markov Chain do not communicate, then they cannot have the same period.

(b) Let P denote the transition probability matrix on a finite state Markov Chain. Then $P^n, n = 2, 3, \ldots$ are stochastic matrices.

(c) A Markov chain on a countably infinite state space necessarily has all states being recurrent.

(d) Communication classes in a Markov chain are closed subsets of the state space.